Dynamic Structural Banking

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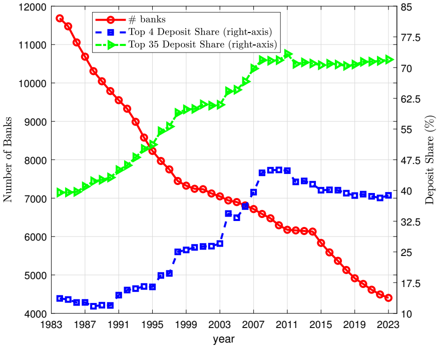
Motivation: Growing Concentration

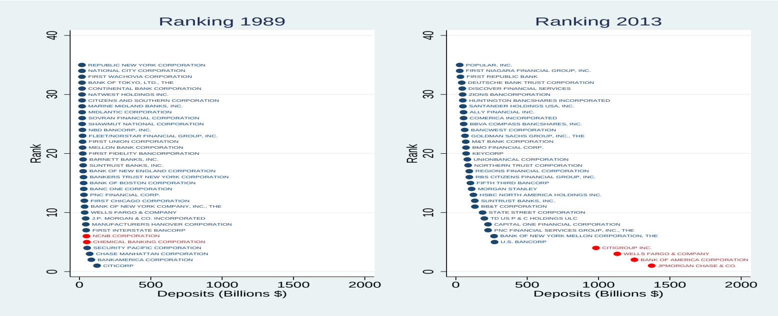
Figure: Concentration Measures and Number of Banks



- Following the Riegle-Neal Regulatory Act in 1994, banking industry concentration has grown considerably as banks branched out via mergers across state lines.

Granularity in the Size Distribution of Banks

Figure: Deposit Distribution Pre- and Post-Reform



Note: Deposits in reported in real terms. Red bubbles identify banks ending in the Top 4 (Chase, BOA, Wells, Citi) of the distribution post-reform (2013). Source: Call Reports.

"I want to be very, very clear: too-big-to-fail is one of the biggest problems we face in this country, and we must take action to eliminate too-big-to-fail."

Ben Bernanke, Time, December 28, 2009/January 4, 2010, p. 78.

Questions

1. A Fixed Point problem for policy makers: Regulation affects market structure and market structure affects regulation.
   - 1.A How has regulation contributed to granularity in the banking industry? That is, spillovers of idiosyncratic shocks to big banks to the aggregate economy?
   - 1.B Has rising concentration mitigated or exacerbated financial instability?

Questions

1. A Fixed Point problem for policy makers: Regulation affects market structure and market structure affects regulation.
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   - 1.B Has rising concentration mitigated or exacerbated financial instability?

2. Methodological Challenges: Integrating macro-finance and industrial organization models with heterogeneity to match the data.
   - 2.A Unlike most macro models with heterogeneity which assume atomless (measure zero) agents, an equilibrium model with big granular banks requires different computational methods.
   - 2.B An idiosyncratic shock to an atomless bank (or a finite set of banks in a continuum economy) has no aggregate effects (i.e. can solve for a steady-state cross-sectional distribution).
   - 2.C An idiosyncratic shock to a systemically important granular bank has aggregate effects generating a sequence of cross-sectional distributions over time requiring different solution methods.

Outline

1. Some Relevant Regulatory History
2. Some Banking Industry Data to Guide Model Selection
3. Static Building Block Theory
   - Maturity Transformation & Liquidity Services ($\longrightarrow$ Deposit Channel)
   - Delegated Monitoring ($\longrightarrow$ Lending Channel)
   - The IO of Banking ($\longrightarrow$ Competition & (in)Stability)
   - Spatial Competition ($\longrightarrow$ Inefficient Entry)
4. A Simple Symmetric Dynamic Model
5. Heterogeneous Bank Models
   - The Deposit Channel
   - Bank Mergers
   - The Lending Channel
   - Putting the Two Channels Together
   - Computational Challenges
6. Directions for Future Research

Today's Literature Review

Imperfect Competition in the...	
Loan Market	Deposit Market
Corbae & D'Erasmo (2021, ECMA; 2025, JPE Macro)	Egan, Hortaçsu, & Matvos (2017, AER)
Wang, Whited, Wu, & Xiao (2022, JF) Aguirregabiria, Clark, Wang (2025, AER)	

A more complete survey of today's lecture is in Corbae "Regulation and Competition: Implications for Efficiency and Stability in the Banking Industry", forthcoming in the Econometric Society World Congress Monograph Series (accessible on my website)

Regulatory Policy: Implications for Models

- Good regulatory banking policy is designed to remedy certain underlying financial frictions, otherwise it may create a friction itself.

- Those policies may also face tradeoffs. For example, bank deposit insurance to prevent non-fundamental runs may create a moral hazard problem inducing excessive risk taking (Keeley (1990, AER).

- Capital requirements to correct excessive risk taking by imposing more "skin-in-the-game" by equity holders may induce regulatory capital arbitrage whereby banks take their risky loans off-balance-sheet and/or shift lending to unregulated (shadow) nonbanks studied by Buchek, etal. (2024, JPE).

Some Regulatory History

- Federal Reserve Act of 1913: Creation of the Fed in response to bank liquidity shocks owing to bank panics (Discount Window/Fed funds).
- Glass-Steagall Act of 1933: Further response to panics creating Deposit Insurance (bank moral hazard concerns).
- Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994: removed cross-state restrictions (competition, efficiency, and stability).
- Gramm-Leach-Bliley Financial Services Modernization Act of 1999: removed barriers between commercial and investment banks (economies of scope).
- Sarbanes-Oxley/OCC Heightened Standards Acts of 2000s: tightened board oversight and transparency of financial information (in response to agency and opacity issues).
- Dodd-Frank Act of 2010: tighter capital and liquidity requirements varying with bank size, G-SIB stress testing, Orderly Resolution of Failing banks (in response to risk taking, spillovers, and failure).

Some Banking Data Facts

In Corbae-D'Erasmo (2025) we provide empirical evidence for:

1. Imperfect Competition (elevated Herfindahls, incomplete passthrough, rising markups).
2. Large right tail deviations from Zipf's law.
3. Geographic expansion across states.
4. Diversification of idiosyncratic deposit shocks.
5. Decreasing average costs across size ($\rightarrow$ Increasing returns from scale and scope).
6. Countercyclical failure rates and size dependent bank entry and exit.
7. The std. dev. of charge-off rates, default frequencies, and loan interest rates cost decline with bank size.
8. Statistically insignificant mean differences across bank size in the cost of deposits, loan interest rates, and the return of (net) securities but we do find statistically significant differences in markups increasing with bank size.
9. Countercylical chargeoffs, net interest margins, and markups.

Data Facts (cont.)

In Corbae-D'Erasmo-Smith (2025) we provide empirical evidence for:

I. Using a granular view (along the lines of Gabaix (2011,ECMA)), we find an idiosyncratic shock to bank deposit flows of G-SIB banks spill over to the aggregate economy via changes in loans, total credit, and GDP.

II. Most mergers are by big banks acquiring small banks (the median acquirer has over seven times more assets than the target) and are procyclical.

III. Tobin's Q rises during the transition between the pre-Riegle-Neal and post-Dodd-Frank periods when $Q-$ratios are approximately at their "steady state" value of 1.

IV. Average county level bank HHI's are 2191 over 1984-1993 and 2575 from 2010-2019 (within anti-trust enforcement of Nocke-Whinston (2022, AER)).

V. Despite rising Herfindahls *within* the banking sector, the growing non-bank sector competes with banks *across* the credit market.
   - Credit to the Private Non-Financial Sector by Banks to Total has fallen from 45% in 1984-1993 to 33% in 2010-2019.

Data: A Simple Balance Sheet

Table: Balance Sheet by Bank Size

	Bank Size			
	All Banks	Top 4	Top 5-35	Fringe
Assets (% of Total Assets)				
Securities, Cash, Fed Funds Loaned (a^+)	35.68	30.43	30.81	35.70
Loans (ℓ)	53.83	61.30	59.96	53.80
Fixed & Intangile Assets (Υ)	10.49	8.27	9.22	10.50
Liabilities (% of Total Assets)				
Deposits (d)	88.38	83.80	83.25	88.41
Fed Funds Borrowed (a^-)	1.14	7.93	7.53	1.11
Equity (e)	10.49	8.27	9.22	10.50

Note: Data corresponds to commercial banks in the US. While we exclude assets and liabilities that are not in our simple models, it accounts for over 95% of assets and liabilities. Source: Consolidated Report of Condition and Income.

- Large banks rely more on other borrowings while small banks resort to insured deposits to fund loans since other external funding is relatively costly (as predicted by corporate finance theories).

Building Block Theory: Why Do Banks Exist?

1. Maturity Transformation/Liquidity Services (Diamond-Dybvig (1983, JPE)):
   - Illiquidity of assets provides the rationale both for the existence of banks and for their vulnerability to runs.
   - Deposit insurance can eliminate the bad eqm., but can generate a moral hazard problem of excessive risk taking.
2. Delegated Monitoring (Diamond (1984, RESTUD)):
   - It is inefficient for many small lenders to incur fixed monitoring costs of large borrowers; instead delegate the monitoring to one lender (a bank).
   - If the bank is sufficiently diversified, it cannot misreport its private information about solvency.

Building Block Theory: The IO of Banking

1. Competition and Stability:
   - Lower profit margins with limited liability induce banks to reach for risky yield in competitive markets causing financial instability (Allen-Gale (2004,JMCB)).
   - Lower loan interest rates in competitive markets can induce positive selection of less risky borrowers causing financial stability (Boyd-DeNicolo (2005,JF)).

2. Competition Across Space:
   - Applying Salop's (1979, Bell) circular model to banking across space with fixed entry costs (scale economies), yields a simple model of monopolistic deposit competition where the decentralized equilibrium exhibits "too much" bank entry/competition relative to the socially optimal number of banks.

A Simple Dynamic Model with Endogenous Bank Market Shares

Corbae-Levine (2025) develop a tractable dynamic industry equilibrium model to assess the impact of regulatory policy on market efficiency and stability.

- Simple idea: More competition is good for market efficiency but low profitability raises incentives to reach for risky yields $\longrightarrow$ financial instability
- Model frictions include:
  - Deposit insurance & limited liability induce moral hazard & excess risk taking
  - Deposit market power imposes a pecuniary externality in external finance
  - Agency frictions between executives and shareholders
- Consider short and long run effects of regulatory policies
- Model is sparsely parameterized.
- Model predictions are consistent with empirical regression results:
  - Increased competition induces increased risk taking
  - Better governance and tighter leverage induces less risk taking
  - There is an interaction effect between governance and leverage policies

Model Environment

- Choice Variables: S,D,N,E, Parameters: A,η,b,γ,δ, Policy: $\kappa, \lambda, \beta, \alpha$
- N banks Cournot compete for insured deposits (free entry gives N).
- Bank i deposits D_i fund bank loans subject to a risk-return tradeoff:
  - For each unit input, portfolio yields $A \cdot S_i$ with probability $p(S_i)$ and yields 0 otherwise where $S_i \in [0,1]$.
  - Higher return S projects are less likely to succeed since $p'(S) < 0$.
  - Parameterize $p(S_i) = 1 - {S_i}^{\eta}$, $\eta \geq 1$; higher $\eta \rightarrow$ better monitoring.
- Inverse deposit supply function $r_D(Z) = b + \gamma Z$, with $Z = \sum_{i=1}^{N} D_i$.
- External borrowing costs from monetary policy parameterized by α.
- Risk neutral shareholders with discount factor δ make the initial equity injection E_i to finance fixed entry costs κ.
- Bank managers choose the riskiness of the loan portfolio S_i and its scale D_i to maximize discounted (at rate $\beta \leq \delta$) bank profits s.t. a leverage constraint that $\frac{D_i}{E_i} \leq \lambda$.

Decision Problems

- Incumbent bank i's static profit imposing limited liability is

$$\pi_i(S_i, D_i; N) = p(S_i) \overbrace{\left[A \cdot S_i - (r_D(Z) + \alpha)\right]}^{\text{interest margin } R_i} D_i \tag{1}$$

where $r_D(Z)$ is the inverse deposit supply function and $Z = \sum_{i=1}^{N} D_i$ is aggregate lending in a market with N banks.

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- An incumbent manager of bank i maximizes the present value of the solvent bank at discount rate β with dynamic problem given by

$$V_i(N) = \max_{S_i, D_i} \pi_i(S_i, D_i; N) + \beta p(S_i) V_i(N') \tag{2}$$

$$\text{subject to} \qquad \frac{D_i}{E_i} \leq \lambda \tag{3}$$

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- Shareholders with discount rate δ will inject equity to fund bank i entry provided

$$E_i\left(N\right) \equiv \frac{\pi_i(N)}{1 - \delta p(S_i)} \geq \kappa. \tag{4}$$

Equilibrium Risk Taking and Lending

- In the short run, optimal bank choices over portfolio risk (S) and external funding (D) in a symmetric nonbinding equilibrium are:

$$p(S) = -\frac{p'(S)}{A} \cdot \left[R + \beta \cdot \frac{E(N)}{D} \cdot w(S)\right], \tag{5}$$

$$R \equiv A \cdot S - (r_D(Z) + \alpha) = \frac{r'_D(Z)}{N} \cdot Z, \tag{6}$$

where the "agency wedge" is

$$w(S) \equiv \frac{[1 - \delta p(S)]}{[1 - \beta p(S)]} \leq 1. \tag{7}$$

- (5) implies that the correlated probability of success $p(S)$ is:
  - Positively related to interest margins (since $-p'(S) > 0$,).
  - Inversely related to leverage and agency conflicts.
  - There is an interaction between leverage and agency absent in a static model.
- (6) implies that the interest margin R is declining in competition (since $\frac{r'_D(Z)}{N} = \frac{\gamma}{N}$) (i.e. more competition, more market efficiency)

Cournot Equilibrium

Taking policy parameters $\Theta = (\kappa, \beta, \alpha, \lambda)$ as given, a *symmetric steady state Cournot equilibrum* is simply 3 equations in 3 unknowns:

- F.O.C. w.r.t. Risk Taking S ($\rightarrow$ loan portfolio success prob. $p(S)$).
- F.O.C. w.r.t. Deposit Funding D ($\rightarrow$ aggregate lending $Z = N \cdot D$).
- Free entry condition N ($\rightarrow$ bank market concentration $\frac{1}{N}$).

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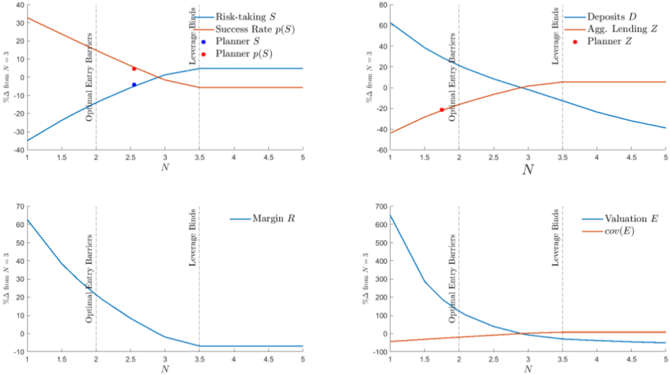
Our experiments (unanticipated policy shocks Θ') consider two cases:

- Short Run: Taking market structure N as given, how do S and Z change transitioning with expected duration to a new long run N' in $20 = \frac{1}{\zeta}$ years as in Fig. 1 (i.e. $\zeta = 0.05$)?

$$V_{\Theta'}(N) = \max_{S,D} \pi_{\Theta'}(N) + \beta p(S_{\Theta'}) \left[(1-\zeta)V_{\Theta'}(N) + \zeta V_{\Theta'}(N')\right]$$

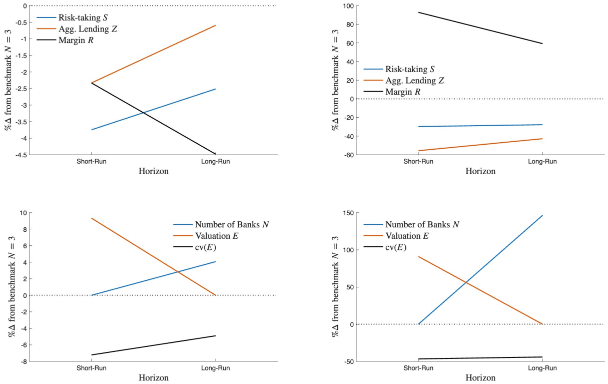
- Long Run: Market structure N' changes since policy affects the charter value of the bank (and hence entry).

Competition, Stability, and Efficiency



- Too little (too much) competition $\longrightarrow$ inefficiently low (high) risk taking.
- Since decentralized bank choices differ from the social planner $\longrightarrow$ role for regulatory policy (raise κ to induce $N = 2.5$ - i.e. implement near duopoly).

Governance & Regulatory Counterfactuals



- Better governance lowers risk taking, raises market efficiency, increases market value in the short run, and competition in the long run.
- Tighter leverage lowers risk taking, reduces market efficiency, increases market value in the short run, and competition in the long run.

Empirical Results - Agency and Leverage

$$\sigma_{ist} = \gamma_C \cdot C_{ist} + \gamma_L L_{ist-1} + \gamma_A A_{ist-1} + \gamma_G G_{ist} + \gamma_I G_{ist} \cdot L_{ist-1} + \theta_i + \theta_{st} + \varepsilon_{ist}$$

Table: On Competition, Governance, and Bank Risk

	Model Prediction	Regression Coefficient
Bank Competition	$\gamma_C > 0$	0.5994*** (0.1778)
Leverage-Lagged	$\gamma_L > 0$	0.0119** (0.0048)
Ln(Bank Assets)-Lagged	$\gamma_A < 0$	-0.1919** (0.0748)
% Institutional Ownership	$\gamma_G < 0$	-1.1725*** (0.1968)
Leverage*Institutional Ownership	$\gamma_I > 0$	0.0497*** (0.0129)
Observations		1994
R-squared		0.7945

σ_{ist} denotes the standard deviation of daily bank i stock returns. The sample consists of BHC-year observations from 1987 through 1995. *, **, and *** indicate significant at 10%, 5%, and 1%, respectively.

Heterogeneity in the Banking Sector

- The previous papers either assumed a "representative" bank or symmetric equilibria.
    - Quantitative Central Bank DSGE models (e.g. Gertler-Karadi (2011, JME)) fall in this category.
- As we saw in the data section, there is substantial heterogeneity in the U.S. banking industry.
- Just as the firm dynamics literature added idiosyncratic shocks to DSGE models to generate endogenous heterogeneity, recent banking models have included, for example, idiosyncratic shocks to deposits to generate an endogenous cross-section with granular banks.
- While there is a body of research on networks that considers financial contagion through balance sheet interconnectedness (eg. Fed Funds), the next 3 sections describe a new body of literature with strategic spillovers in the deposit or loan markets arising from granular banks.

A Quantitative Model of Bank Runs

- Egan-Hortascu-Matvos (2017, AER) provide a heterogeneous bank version of Diamond-Dybvig to address the quantitative significance of spillovers between G-SIB banks in the presence of fears about insolvency.
- Consumer preferences are modeled as in the IO, discrete choice literature over returns from insured and uninsured bank deposits while bank "loans" yield an exogenous stochastic return.
- Since banks can have negative cash flows and there is limited liability, equity holders must decide whether to inject funds to retain a bank's charter value.
- K granular banks Bertrand compete over returns on insured and uninsured deposits and choose whether to default. Rational depositors factor each bank's default probability into their bank choice.
- Uninsured depositor utility depends on bank survival and bank survival depends on uninsured depositor demand. This can lead to multiple eqa.

EHM Bank Bellman Equation

- The Bellman equation for bank k equity value is

$$E_k = \max_{i_k^I, i_k^N} \int_{\bar{R}_k}^{\infty} \Big[M^I s_k^I(i_k^I, \mathbf{i}_{-k}^I)(R_k - c_k - i_k^I) + M^N s_k^N(i_k^N, \mathbf{i}_{-k}^N, \rho_k, \boldsymbol{\rho}_{-k}^N)(R_k - i_k^N) - b_k + \frac{1}{1+r} E_k' \Big] dF(R_k) \tag{8}$$

where the optimal default threshold $\overline{R}_k$ satisfies (9) with equality

$$\pi_{k,t} - b_k + \frac{1}{1+r} E_k' \leq 0, \tag{9}$$

given debt b_k, the exogenous stochastic return on loans R_k, and (s^I, s^N) are endogenous insured and uninsured deposit market shares.

- Assumptions to turn this into a sequence of static interest rate setting problems and guarantee stationarity (so that $E_k' = E_k$) greatly simplifies computing an equilibrium.

Bank Run Experiment

- After calibrating their model parameters to U.S. data assuming a fundamental eqm., EHM run experiments where they exogenously decrease depositor beliefs about a bank's survival probability that induces a decrease in demand for uninsured deposits, lowering the bank's profitability and increasing its probability of default.

- Since granular banks compete in the deposit market, beliefs have to be consistent with the optimal interest-setting and default behavior of all K banks.

- Hence an idiosyncratic shock to beliefs about bank k's likelihood of failure can spill over to other banks through equilibrium deposit competition effects:
  - to help ease deposit withdrawals at k given beliefs it will be insolvent, k raises its returns on insured and uninsured deposits,
  - due to imperfect competition, other banks need to offer higher deposit interest rates leading to higher overall default rates.

Granular Bank Run Spillover

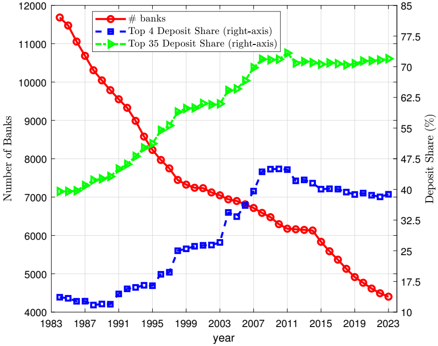
Table: EMH Table 4: Multiple Equilibria 2008 (Percent)

	JPMorgan Chase		Bank of America		Wells Fargo		Citibank		Wachovia	
	Obs.	Eq. 5	Obs.	Eq. 5	Obs.	Eq. 5	Obs.	Eq. 5	Obs.	Eq. 5
Insured interest rate	1.73	2.44	1.98	2.14	2.13	3.04	2.23	2.98	2.08	8.75
Uninsured interest rate	1.73	2.39	1.97	1.96	2.32	3.20	2.23	2.92	2.23	14.06
Probability of default	1.50	2.83	1.82	1.94	1.50	3.49	2.11	3.33	3.28	**52.08**
Insured deposit share	3.38	1.30	9.26	2.57	3.99	1.73	2.07	0.82	5.81	74.47
Uninsured deposit share	15.86	16.06	9.23	9.75	4.30	4.16	16.80	17.33	4.74	0.08

- Obs. column is from calibrated model.
- Eq.5 column is from an equilibrium with an idiosyncratic shock to beliefs that Wachovia's probability of insolvency rises from 3% to 52%.
- In response, uninsured interest rates at Wachovia rise from 2.2% to 14% which induces it to raise insured rates rise from 2% to nearly 9%.
- To compete for deposits, the other big banks must raise their interest rates, cutting profit margins which spills over to a higher probability of default for all banks.

How did we get here?

Figure: Concentration Measures and Number of Banks



- Transition dynamics between two steady states (pre-Riegle-Neal and post-crisis) fueled by merger wave.

A Quantitative Model of Bank Mergers

- Is the US better off as a consequence of relaxing regulations on cross-state branching? What are the costs and benefits?
  - costs: rising markups
  - benefits: exploit increasing returns
- Corbae-D'Erasmo-Smith (2025) build a quantitative model of bank mergers extending the dominant-fringe framework of Gowrishankaran-Holmes (2004, RAND) that endogenously matches the transition path for concentration in Figure 1 and use it to answer those questions.
  - merger wave ends when the gains to deposit growth become smaller than the cost of buying up fringe banks whose charter value is growing as their numbers dwindle.
- Model can be used to explore optimal merger regulations along the lines of Mermelstein, et. al. (2020, JPE) for the banking industry.

Merger Environment

- Granular dominant bank makes TIOLI offer to buy a measure of fringe (measure zero atomless) banks at price determined by fringe banks' charter values.
- Following merger stage, dominant bank makes strategic asset portfolio choice (loans vs. securities) followed by portfolio choices of fringe banks.
- Stackleberg Cournot game in bank loan market determines loan interest rate which competes with a non-bank (shadow) sector in order to match that falling share of bank to total credit market shares pre-Riegle-Neal (45%) to post-2008-crisis (33%).
- Model includes entry/exit decisions critical to evaluate financial stability.
- Model nests standard competitive industry model (aka Hopenhayn) if set entry costs for dominant banks sufficiently high.

Merger Policy Experiment

- After calibrating the model , one can use it to perform a counterfactual asking what would have happened if cross-state branching restrictions had not been lifted in Riegle-Neal?

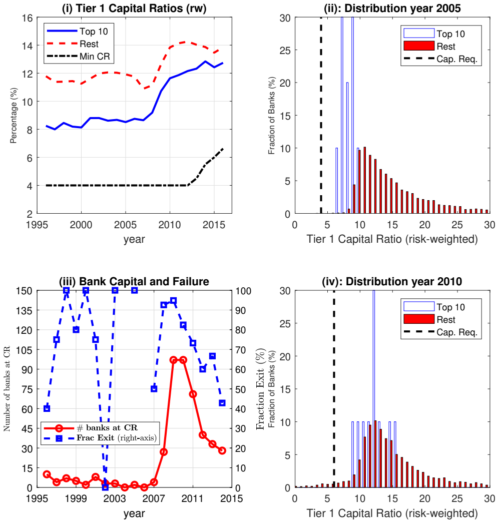
- The counterfactual with branching restrictions assumes a new long run equilibrium of the banking industry with higher costs from Dodd-Frank and technological improvements in the non-bank sector.

- We find a competition/stability tradeoff.
  - Without mergers, the dominant bank is much smaller, competition from the fringe is much more substantial, and measures of competition (such as net interest margin and markups) are much lower.
  - These improvements from competition contribute to a more fragile banking system; the failure rate of fringe banks increases from 0.11% to 0.49% and the value of the banking sector (Tobin's Q) declines substantially.
  - Allocative efficiency improves (using an Olley-Pakes (1996, ECMA) decomposition) as mergers help exploit economies of scale driven by fixed costs.

The Bank Lending Channel

- Previous sections focused on the deposit side of the balance sheet that has been studied extensively in the IO literature on banking given detailed data.
- Macro-finance papers (e.g. Bernanke-Gertler (1995, JEP)) focus on the credit channel of monetary policy in 2 parts.
- Heterogeneity matters:
  - Balance Sheet Channel: liquidity constrained nonfinancial firms find it costly to borrow and invest following monetary tightening (e.g. Ottonello-Winberry (2020, ECMA))
  - Lending Channel: liquidity constrained banks find it costly to fund lending following monetary tightening (e.g. Kashyap-Stein (2000, AER))).
- Corbae-D'Erasmo (2021, ECMA and 2025, JPE Macro) build heterogeneous banking models consistent with the bank lending channel to evaluate the role of capital regulation and too-big-to-fail.

The Crisis and Basel III Capital Regulation



Essentials of Lending Environment

- Focusing on the lending channel, CD estimate a panel AR1 for bank specific idiosyncratic deposit shocks and find small bank variances significantly exceed that of big banks (consistent with geographic diversification).
- Dynamic Dominant-Fringe model generates strategic spillovers consistent with the granularity literature and nests an industry with perfect competition.
  - Idiosyncratic shocks to granular banks generates aggregate uncertainty which requires computational methods to approximate strategic equilibria like that in Ifrach and Weintraub (2017, RESTUD).
- Borrowers make a discrete choice over bank or nonbank loans and a continuous risk-return choice that endogenizes bank portfolio risk.
- Banks bear costs to monitor borrower projects generating increasing returns as in Diamond (1983, RESTUD).
- Bank entry and exit decisions endogenize the bank size distribution.

CD Bank Bellman Equation

The Bellman equation for a bank of type τ solves

$$(\mathcal{T} v_\tau)(s,S) = \max_{a_\tau \in A_\tau(s,S)} \mathbb{E}_{(s',S')|(s,S)} \left[\max_{x'_\tau \in \{0,1\}} (1-x'_\tau)\big(\mathcal{D}_\tau(a_\tau; a^{\neq}, s, S, s', S') + \beta v_\tau(s',S')\big) \right]$$

s.t.

$$\mathcal{L}_B \equiv \int \ell_d(s,S)\mu_d(ds,S) + \int \ell_f(\ell_d, \ell_d^{\neq}, s, S)\mu_f(ds,S) = \mathcal{L}_B^D(S, r_B^L, r_N^L), \quad (10)$$

$$(\mathcal{T}^* \mu_\tau)(s_0,S_0) = \int (1 - x'_\tau(a_\tau; a^{\neq}, s, S))\mathbf{1}_{\{a_\tau \in s_0\}} G(S_0|S)\mu_\tau(ds,S) + M'_\tau \overline{G}(s_0,S_0). \quad (11)$$

where (s,S) denote individual bank and aggregate states.

- Loan Market Clearing in (10): dominant bank internalizes how its loan choice affects fringe loan decision rules $\ell_f(\cdot)$ and the bank loan interest rate r_B^L recognizing the competition from nonbanks offering loans at r_N^L.
- The law of motion for the cross-sectional distribution of banks μ in (11) ensures that a dominant bank internalizes its impact on future market structure.
- A Markov Perfect Cournot Equilibrium requires solving a big fixed point problem defined by the two operators $\mathcal{T}$ and $\mathcal{T}^*$ and market clearing.

The Marginal Propensity to Lend

- The foc for $\ell_\tau \in A_\tau(\Sigma)$ where $\Sigma = (s, S)$ trades off benefits and costs:

$$\begin{aligned}\theta_\tau(\Sigma) &= \mathbb{E}_{\Sigma'|\Sigma}\left\{[p(\Sigma')r_B^L - (1-p(\Sigma')\lambda'] + \left[p(\Sigma')\frac{\partial r_B^L}{\partial \mathscr{L}_B}\frac{\partial \mathscr{L}_B}{\partial \ell_\tau}\ell_\tau\right] + \left[\beta\frac{\partial v_\tau(\Sigma')}{\partial \Sigma'}\frac{\partial \Sigma'}{\partial \ell_\tau}\right]\right\} \\ &- \mathbb{E}_{\Sigma'|\Sigma}\left\{\left[\frac{\partial Pr(\pi_\tau(\Sigma') < 0, x'_\tau = 0))}{\partial \ell_\tau}\frac{\partial \psi_\tau(|\pi_\tau(\Sigma')|)}{\partial |\pi_\tau(\Sigma')|}\right] + r^A + \frac{\partial C_\tau(\ell_\tau)}{\partial \ell_\tau}\right\}\end{aligned}$$

where θ_τ is the multiplier on the b-o-p external financing constraint

- $\theta_\tau(\Sigma)$ determines whether MPL is either 0 or 1 across different states Σ.
- In some states where deposit realizations are low, banks behave like "hand-to-mouth" consumers.
- The 2nd and 3rd terms are absent for fringe banks, but internalized by granular, dominant banks.
- Differences in equity finance costs $\psi_\tau(\cdot)$ and loan monitoring costs $C_\tau(\cdot)$ affect the MPL

Putting the Model to Work

- After estimating model parameters by SMM of our Markov Perfect Cournot Equilibrium, validate it by showing simulated data from our model is consistent with:
  - The bank lending channel; small banks lower lending more in response to a rise in funding costs as in Kashyap and Stein (2000).
  - Competition/Stability Hypotheses; probability of a crisis is decreasing in concentration and default frequencies are increasing in concentration.
  - Granularity; idiosyncratic deposit shocks to dominant G-SIB banks can explain up to 34% of bank intermediated output and 1% of total output.

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- Use the model to address:
  - the role of capital regulation on markups, allocative efficiency, and stability in Corbae-D'Erasmo (2021, ECMA).
  - the role of too-big-to-fail (TBTF) on risk taking and aggregate lending in Corbae-D'Erasmo (2025).

Capital Regulation

- How much do higher Basel III size and countercyclical capital requirements affect lending, financial stability, market structure (both within and across credit (bank/nonbank) sectors), allocational efficiency, and welfare?
- Big banks raise equity issuance and lower dividends to fund lending while small banks lower lending and unprofitable ones exit (selection effects).
- Exit increases in SR and decreases in LR. Entry falls in both SR and LR → more concentrated banking industry (fringe bank market share falls) with higher markups at expense of rising unregulated nonbank market share in total credit supply.
- Allocative efficiency (Olley-Pakes (1996, ECMA)) rises with the regulatory reform, consistent with empirical studies (e.g. Berger-Hannan (1998, RESTAT) of the rise in concentration on efficiency.
- Short run welfare losses but long run welfare gains.

Putting the two sides together

- EHM (2017) endogenize deposit market competition but have exogenous stochastic loan returns. CD (2021,2025) endogenize loan market competition but have exogenous stochastic deposit market inflows.
  - CDS(2026) endogenize both deposit and loan market structure through mergers and endogenous entry/exit but take the deposit rate r^D as given.

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  - Importantly, they find that bank market power explains much of the transmission of monetary policy to borrowers, with an effect comparable to that of bank capital regulation.
- Aguirregabiria, Clark, Wang (2025, AER): estimate a static structural model of bank oligopoly competition for both deposits and loans in multiple geographic markets in order to assess imbalances of credit flows.
  - Importantly, nonbanks do not share economies of scope of taking deposits *and* making loans; ACW find that an increase in nonbank competition in the loan market can spill over to less bank deposit taking.

Computational Challenges

- In general, solving models with heterogeneity involves application of two fundamental operators along with market clearing in a nested fixed point problem:
  - $\mathcal{T}$: mapping value functions in the space of continuous bounded functions to continuous bounded functions. This contraction operator is straightforward (Blackwell's Theorem provides simple conditions for existence & uniqueness).
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- Models with granular, strategic heterogeneous banks face a curse of dimensionality depending on the state space:
  - Even if an entry condition treats all banks as identical, ex-post idiosyncratic shocks create aggregate uncertainty generating an infinite sequence of cross-sectional distributions $\{\mu_t\}_{t=0}^{\infty}$ (that's the idea behind granularity absent in perfectly competitive models).
  - Hence, approximation methods must often be applied. For instance, CD (2021, ECMA) and WWWX (2022, JF) apply methods along the lines of Ifrach and Weintraub (2017, RESTUD) which approximate the cross-sectional distribution with a finite set of moments.

Directions for Future Research

- Economies of Scope: While much of the previous sections have focused on how Riegle-Neal allowed banks to exploit economies of scale and diversification, little focus was placed on economics of scope that were made possible by the Gramm–Leach–Bliley Act of 1999. What accounts for different bank business models?

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- Complexity: Growth in the geographic span and scope of big banks can be an important issue for systemic risk, and is taken up empirically in Correa-Goldberg (2022, JB&F) by constructing measures of complexity and testing how they affect risk taking.
  - Future research using quantitative models to understand the tradeoffs in bank opacity along the lines of the simple 3 period model in Dang, et.al. (2017, AER) to governance and supervision by regulators.
  - There's some deep theory on information design and private information problems to be addressed in this literature.

Directions for Future Research (cont.)

- Global Banking: A related agenda which shares the objectives of understanding geographic expansion and complexity is that of global banking. As stated (p.167) in Buch-Goldberg (2020, ARFE):
  - "...the response of global banks to shocks is highly heterogeneous...
  - Responses to shocks are larger when bank-specific capabilities to absorb shocks are weaker, when bank capital is lower, and when bank internal liquidity management is less effective.
  - Responses also depend on the business model of the parent bank, the importance of the foreign market for the overall balance sheet, and the degree of its geographic or organizational complexity.
  - Country specific macroeconomic factors matter as well, including the policy environment...This is the point at which micro- and macroeconomic views need to be combined."

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- Optimal Regulation: Many of the quantitative papers computing optimal policy take market structure as exogenously given.
  - It is critical to evaluate how regulation affects market structure in the long run which has implications for welfare and allocational efficiency.
  - How far is a decentralized optimal allocation to a constrained efficient allocation with deep financial frictions?

Appendix

Cross-Section of Rising Bank Concentration

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- How did they get there?
- From 1989 to 1992, North Carolina National Bank acquired over 200 (small) banks and in 1992 changed its name to Nationsbank.
- In 1996 Chemical bought Chase but kept the Chase name.
- In 1997 Norwest bought Wells Fargo (and took their name) and Nationsbank bought BoA (and took their name).
- In 2000 Chase bought JP Morgan (and added their name)

Return